Starter

Sketch and state the domain and range:

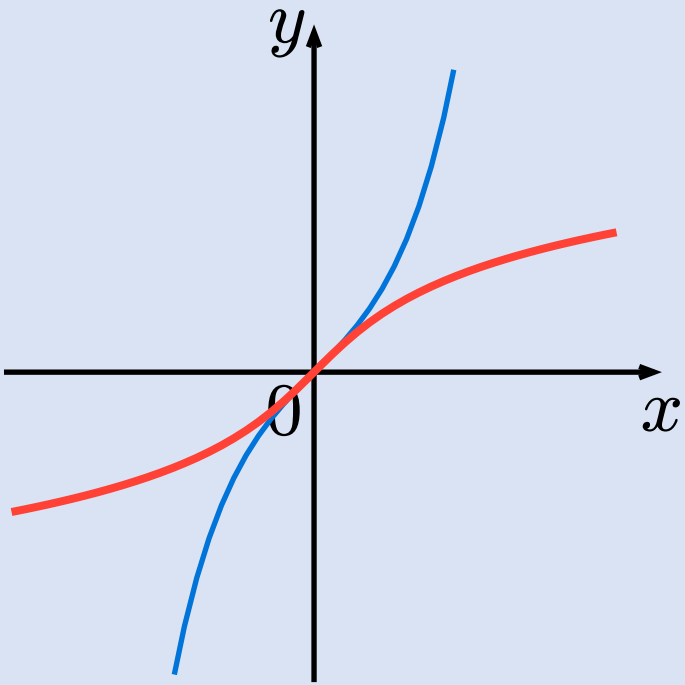
1. $y = \sinh(x)$
2. $y = \cosh(x)$
3. $y = \tanh(x)$
4. $y = \operatorname{arsinh}(x)$
5. $y = \operatorname{arcosh}(x)$
6. $y = \operatorname{artanh}(x)$

$$y = \operatorname{arsinh}(x)$$

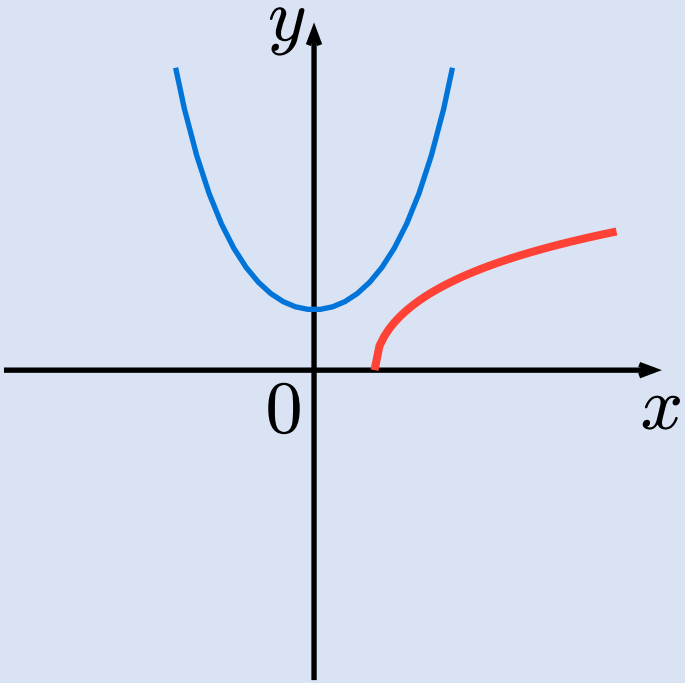
Domain: $x \in \mathbb{R}$

Range: $y \in \mathbb{R}$

Odd function



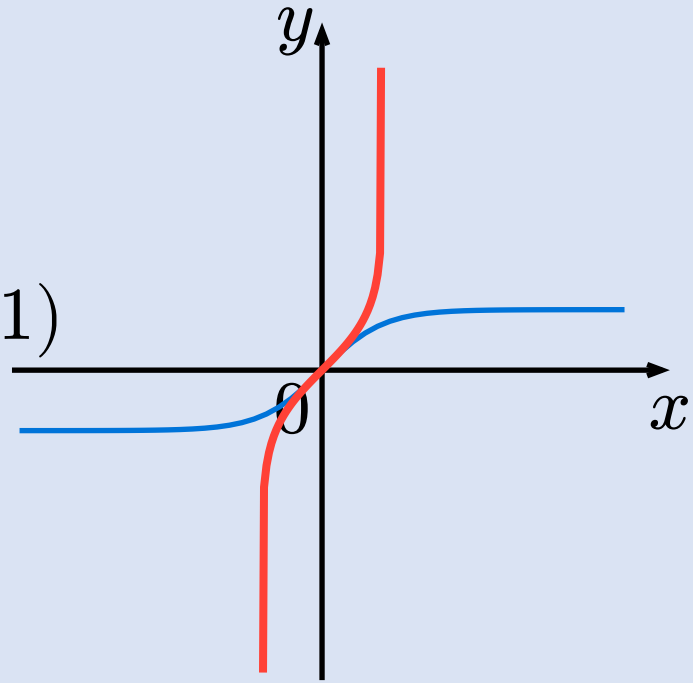
$$y = \operatorname{arcosh}(x)$$



Domain: $x \geq 1$

Range: $y \geq 0$

$$y = \operatorname{artanh}(x)$$



Domain: $x \in (-1, 1)$

Range: $y \in \mathbb{R}$

Odd function

Show that

$$\operatorname{arsinh}(x) = \ln\left(x + \sqrt{x^2 + 1}\right)$$

You Try

1. Write $\operatorname{arsinh} 4$ in the form $\ln\left(a+\sqrt{b}\right)$ where $a, b \in \mathbb{Z}$.
2. Write $\operatorname{arsinh}(-4)$ in the form $\ln\left(a+\sqrt{b}\right)$ where $a, b \in \mathbb{Z}$.
3. Write $\operatorname{arcosh} 5$ in the form $\ln\left(a+\sqrt{b}\right)$ where $a, b \in \mathbb{Z}$.
4. Write $\operatorname{artanh} \frac{3}{4}$ in the form $\ln(a)$ where $a \in \mathbb{R}$.
5. It is given that $\operatorname{cosech} x = \frac{3}{4}$
   - Find the exact values of $\sinh x$ and $\cosh x$.
   - Hence find the exact value of x, in terms of natural logarithms.

Q5

It is given that $\operatorname{cosech} x = \frac{3}{4}$

- Find the exact values of $\sinh x$ and $\cosh x$.
- Hence find the exact value of x, in terms of natural logarithms.

$$\operatorname{cosech}\ x = \frac{3}{4} \Rightarrow \boxed{\sinh x = \frac{4}{3}}$$

$$\cosh^2 x - \sinh^2 x = 1 \Rightarrow \cosh^2 x = 1 + \frac{16}{9} \Rightarrow \boxed{\cosh x = \frac{5}{3}}$$

$$\sinh x = \frac{4}{3} \Rightarrow x = \operatorname{arsinh}\frac{4}{3} = \ln\left(\frac{4}{3} + \sqrt{\frac{16}{9} + 1}\right) = \boxed{\ln 3}$$

Solve

$$\sinh^2 x - 4\cosh x + 5 = 0$$

You Try

1. $3\sinh x + 2\cosh x = 4$
2. $4\cosh x - 8\sinh x + 1 = 0$
3. $\cosh x + 4\sinh x = 3$
4. $3\operatorname{sech} x - 2 = 5\tanh x$
5. $9\cosh^2 x - 6\sinh x = 17$
6. $16\sinh x - 9\cosh x + 6\sinh 2x - 12 = 0$

Evaluate without a calculator

$$\cosh(\ln 5)$$

Evaluate without a calculator

$$\cosh(\ln 5)$$

$$\frac{13}{5}$$

Evaluate without a calculator

$$\sinh(3\ln 2)$$

Evaluate without a calculator

$$\sinh(3\ln 2)$$

$$\frac{63}{16}$$

Evaluate without a calculator

$$\cosh\left(\frac{3}{2}\ln 4\right)$$

Evaluate without a calculator

$$\cosh\left(\frac{3}{2}\ln 4\right)$$

$$\frac{65}{16}$$

Evaluate without a calculator

$$\tanh(\ln 3)$$

Evaluate without a calculator

$$\tanh(\ln 3)$$

$$\frac{4}{5}$$

Practice

Write $\cosh^3 x$ in the form $a \cosh 3x + b \cosh x + c$ where $a, b, c \in \mathbb{R}$

Similarly, find $\sinh^3 x$ in terms of $\sinh 3x$ and $\sinh x$.

Practice

TLMaths FM Core Pure Textbook Part 2.

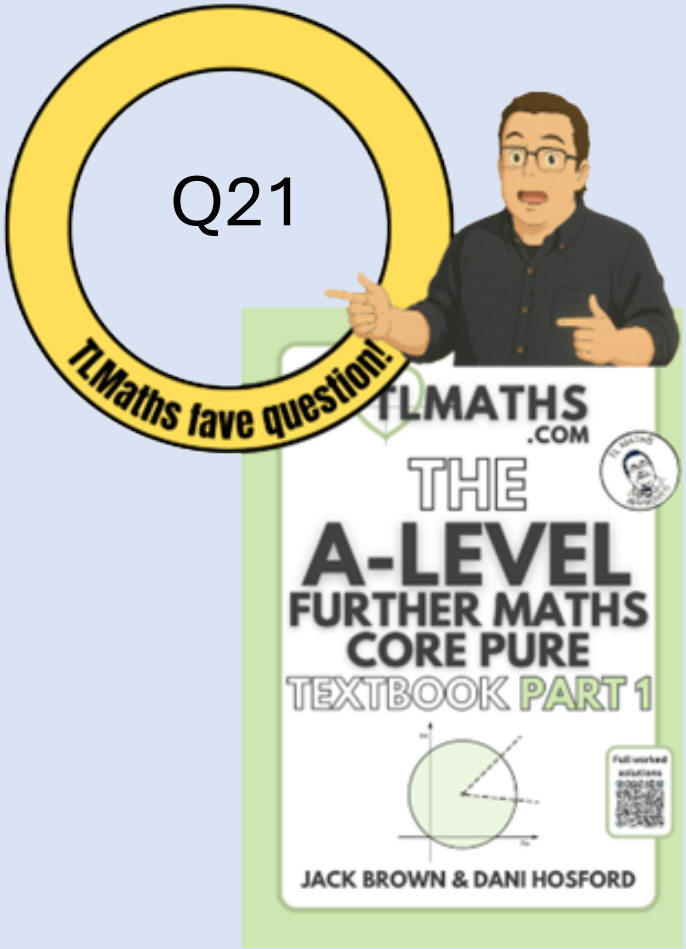
Hyperbolic Graphs Practice:

Hyperbolic Functions: Domain and Range

Q1-7

Hyperbolics Identities:

Hyperbolic Functions: Osborn's Rule and Proof

Q1-4 and 8-10 , Q12, 14, 16, 17, 19, 21, 25, 26

Q28, 29, 31, 32, 35, 37, 41, 42, 43